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Development of process safety knowledge graph: A Case study on delayed coking process

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ABSTRACT

Process safety is one of the essential preconditions for the achievement of green manufacturing. The improvement of process safety management requires a comprehensive risk analysis based on the collection of almost all safety related information, which are usually unstructured knowledge and experience. To handle the information and support the risk analysis, a process safety knowledge graph is prompted and the development of domain ontology on delayed coking process is elaborated. The combined top-down and bottom-up approaches are used in defining the process safety schema on ontology level. Several multi-structured data sources are introduced in establishing the process safety knowledge, in which the hazard and operability analysis (HAZOP) reports and process diagrams are most important. The ontology design and data extraction are demonstrated in the manuscript while various related applications are discussed. This process safety knowledge graph might empower the knowledge-based analysis abilities in discovering the hidden relationships between possible risk causes and consequences in an emergency situation, and could provide a foundation for more application related to process safety.

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1. Introduction

As the basic industry of the national economy, in recent years, the chemical industry in China has developed rapidly. Due to stricter safety policies, environmental policies and public appeal all over the world, green manufacturing, which defines the environmentally-friendly operations and process safety (Qian et al., 2017; Yuan et al., 2017; Zhou et al., 2018; Cernansky, 2015), has been increasingly attractive. The achievement of green manufacturing in the chemical industry is more urgent because of the frequent alarms in the production process and recent major accidents (Tauseef et al., 2011; Huang and Zhang, 2015; Wang et al., 2018). To improve the process safety (Crowl and Louvar, 2001), models of different production processes in the chemical industry are the foundation. However, due to the complex relations between different types of equipment, chemicals, process indexes, and accidents in the production processes of the chemical industry, it is extremely difficult to accurately model these production processes. Thus, though a few data-based methods have been proposed to approximate some practical production processes, the modeling of the process safety is still not fully achieved due to the incomplete application of knowledge and information in these processes (Zahedi et al., 2009; Chen and Wang, 2010; Tian et al., 2012).

To improve the process safety, there are several system-based hazard analysis methods promisingly used to derive potential accident scenarios. such as Functional Resonance Analysis Method (FRAM) and Process Resilience Analysis Framework (PRAF). FRAM was first proposed by Hollnagel in 2012 (Hollnagel, 2012) and focuses on the understanding of interactions and emergence phenomena in complex sociotechnical systems. In recent year, FRAM has been applied in oil process units to identify emergent risks (Shirali et al., 2013). In order to incorporate both technical and social factors into a comprehensive risk analysis, Jain et al. have proposed PRAF. PRAF emphasizes dynamic, unpredictable or even unknown types of threats, uncertainty, system degradation and complex interactions. In Jain et al. (2019b) and Jain et al. (2019a), PRAF is applied in cooling tower operations and a batch reactor to improve risk and safety management, respectively. Although these two system-based hazard analysis methods can effectively improve risk and safety management, they all need require quantitative relationship data between various units in the process industry as support, and cannot characterize and use the empirical quantitative relationship in the actual process, which makes the modeling

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of the process incomplete. In addition, due to the inability in automatical construction, the implementation and subsequent improvement of these two methods require a lot of human resources.

As a new powerful tool of knowledge representation, knowledge graph was first proposed by Google in 2012 Steiner et al. (2012). Knowledge graph (Wang et al., 2014; Jia et al., 2018) focuses on describing various entities and concepts in the real world. The knowledge graph can be regarded as a new semantic web, where the ontology is extended at the entity level. With this special form, numerous applications in different fields can be realized with knowledge graph technologies (Qiao et al., 2016), including the semantic search, intelligent question-answering, and intelligent recommendation. Fu et al. extract the public business transaction information of some chosen companies and summarize information of major transaction news on the Internet to build an enterprise information knowledge graph (Chen et al., 2018). Based on this developed knowledge graph, the statistical characteristics of financial laws are tried to be analyzed. Similarly, Bean et al. construct a knowledge graph containing four types of nodes: drugs, protein targets, indications, and adverse reactions (Bean et al., 2017). Based on this graph, the prediction for adverse reactions that were not observed during randomized trials is realized by a machine learning algorithm to improve patient safety. These successful applications in other industries indicate that the knowledge graph technology is a potential solution to promote the process safety by making full use of knowledge and information in the production process of the chemical industry.

A semi-automatic knowledge graph development solution for process safety in the chemical industry is designed in this paper. Each part of the entire development process can be run programmatically with manual supervision and modification. The main contributions of this paper are listed as follows:

(1) Different from FRAM and PRAF, a semi-automatic knowledge graph development solution for improved process safety is proposed in this paper, which can qualitatively and comprehensively model various processes in the form of semantic networks. This construction framework makes full use of the knowledge of process safety in the chemical industry and provides the foundation for improving process safety level. And it can be promoted to other production processes in the chemical industry.
(2) We construct a schema representing the process technologies and concepts in the chemical industry, which is the foundation of the knowledge graph. The main challenge of constructing this schema is how to properly define classes, relations, and properties to describe the process safety analysis process, while minimizing ambiguity and ensuring inference.
(3) To semi-automatically extract useful data from different data sources, such as the existing relational databases and text documents, different extraction solutions are designed based on natural language processing (NLP) technologies.
(4) Based on constructed knowledge graphs, several useful applications, including knowledge visualization, automatic information retrieval and extraction of causal chains, are realized in the case of delayed coking process.

2. Methodology

Ontology is a semantic data model used to define the types of things that exist in our domain and the attributes that can be used to describe them. Ontologies are generalized data models, which means that they only model general types of things that share certain attributes, but not specific individuals. As shown in Fig. 1, 'City' and 'Country' are two generalized models for real cities and countries in real word, thus they can be defined as two concept in the ontology. Using the constructed ontology as a framework, we can add real data, which belongs to different concepts, to create a knowledge graph. Thus, knowledge graph is a collection of a large number of specific individuals according to the constructed ontology. As depicted in Fig. 1, a part of a knowledge graph, including entities 'P.R. China' and 'Shanghai', is constructed based on the constructed ontology.

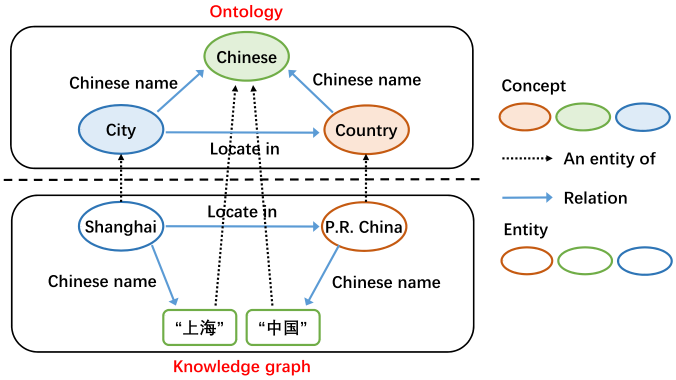


Fig. 1. An example for a part of a knowledge graph and its ontology.

2.1. Ontology design

In most general knowledge graphs, including DBpedia[1], YAGO[2], and OpenKG[3], the encyclopedia data are the main data used to construct knowledge graphs (Wang et al., 2008). Thus, schemas in these knowledge graphs are built in a bottom-up manner (Meilicke et al., 2019) to cover vast amounts of multi-domain data. However, for some special knowledge graphs, the schema can be worked out in a top-down manner (Sohoglu et al., 2012) because it can be extracted by experts easily. The application of both methods must ensure that they do not lead to conflicts at the conceptual level and loss of reasoning. The difference between the above two methods is mainly at the data level. Thus, to choose which method to design the schema for the process safety orientated knowledge graph, the first step is to identify detailed requirements and collect as much relevant data as possible. Herein, several relational databases and some document data, including HAZOP (Hazard and Operability Analysis) (Kotek and Tabas, 2012), operational procedures and process technical specifications, are always collected. For process engineering in the chemical industry, some concepts, such as equipment and chemicals, have been well summarized by experts in the constructed ontology Ontocape (Morbach et al., 2007; 2009), which can describe some factors in process safety analysis without ambiguities and conflicts. Thus, similar concepts can be defined in a top-down manner. However, for concepts related to process safety in HAZOP reports, such as deviations, the top-down method is not suitable because it can no longer guarantee to minimize ambiguity and ensuring inference. Moreover, it cannot ensure that all available information is covered. Thus, for these concepts, the bottom-up method is more desirable. Based on the above analysis, a mixed manner is adopted to define the schema of process safety in this paper. In this mixed method, as many ontologies as possible are first defined by experts in a top-down manner. Then, from the collected data, other ontologies are extracted in a bottom-up manner. Of course, all obtained ontologies need to be manually checked and modified.

[1] https://wiki.dbpedia.org.
[2] https://github.com/yago-naga/yago3.
[3] http://www.openkg.cn/.

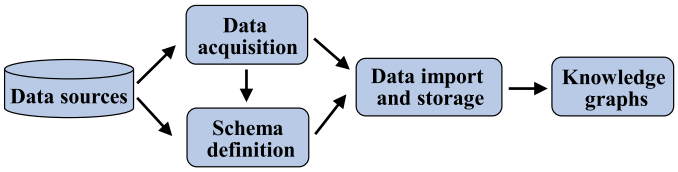


Fig. 2. Construction process of the knowledge graph.

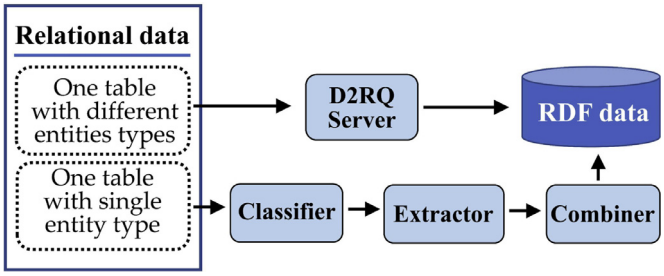


Fig. 3. A solution for data acquisition from relational databases.

2.2. Knowledge graph

In this paper, the main purpose is to construct a process safety knowledge graph for the chemical industry. The general idea of knowledge graph construction is shown in Fig. 2. there are three main components in the construction of process safety knowledge graph, i.e. schema definition, data acquisition, and data import and storage. In the last subsection, the first step, schema definition, has been finished. With the defined schema for process safety, the details of the other two steps are presented in this subsection.

2.2.1. Data acquisition

The knowledge graph is a powerful technique for knowledge representation, whose foundation is a large amount of reliable data. For the process safety knowledge in the chemical industry, there are typically three sources of data, namely the relational databases, human-readable documents and the Internet. The main difference between data in relational databases and documents is whether the data is structured. Considering the quality and reliability of data, the Internet is usually treated as a supplementary data source. For these three kinds of data sources, there are different solutions.

(1) Data acquisition from relational databases

For typical structured data, two solutions are available in different situations. For the original data tables, there are three classical steps, namely table splitting, basic transformation by D2RQ, and post-processing, to transform relational databases to RDFs (Resource Description Framework schema), which is a common data type utilized in the knowledge graph. However, D2RQ is unable to transform data tables where there are multiple entities and relations or one table column involves different entities types. For these data tables, we need to design a new solution, which is shown in Fig. 3. There are three main steps in this new relational data acquisition solution. First, according to the numbers of entities and relations, data tables need to be classified. Next, according to the data table type, we design corresponding programs that extract multiple entities and relationships by iterating through each line of the file with the header field as the dictionary key value. Finally, all extracted data from different kinds of data tables is merged into RDF data on the basis of entity names.

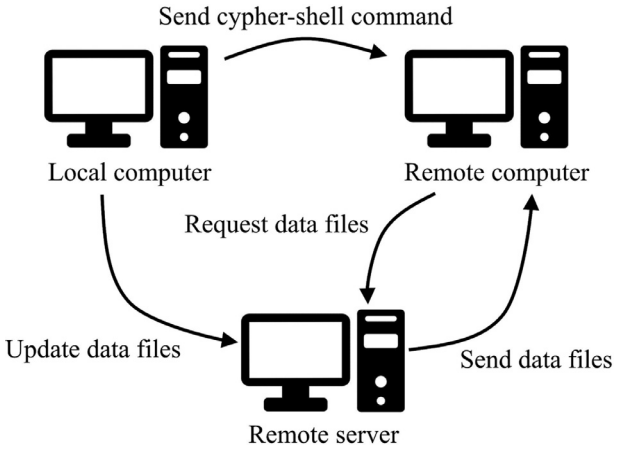


Fig. 4. Realization of remotely importing data into Neo4j.

(2) Data acquisition from documents

Data in human-readable documents, such as HAZOP reports, technical references, and SOPs (standard Operating Procedures), are unstructured and usually hidden in one or several sentences. For these documents, there is no technique can directly acquire data for the knowledge graph. It remains challenging for computers to identify entities and relations hidden in sentences. In this paper, we adopt natural language technologies to overcome this challenge, and the main idea to acquire data from documents is: the entities and relations are identified by NER (Named Entity Recognition) and RE (Relation Extraction) technologies, respectively. After the above two steps, entities and relations are obtained, which need to be checked manually to ensure accuracy. In the case of delayed coking process, 5615 entities and 2680 relations are extracted from HAZOP reports and other documents.

2.2.2. Data import and storage

With the acquired data, the last step to construct a process safety knowledge graph is import and storage of the data. Now, there are several mature and efficient graph database products, including Neo4j and Jena. In this paper, Neo4j, a free high-performance NoSQL graphics database, is chosen as the data storage for process safety knowledge graph because of its extensive practical applications. Considering the need of remote operations in practical applications, it is necessary to implement the function of remotely importing data into Neo4j. The implementation idea is shown in Fig. 4 and described as follows: remote cypher-shell call is realized through neo4j remote listening port, and remote data import is realized by cooperating with data file reading hanging on the server. In the process of remotely importing data, we also implement the function to remove duplicate data, which may make the database chaotic and bring difficulties to the implementation of the query function.

3. Case study

In the above section, the main construction process of process safety orientated knowledge graphs for the chemical industry is proposed. In this section, delayed coking, which is an important process in the oil refining industry, is chosen as a case to develop a knowledge graph to improve the process safety level. The construction process of process safety orientated knowledge graphs for delayed coking is shown in Fig. 5. Some details in this case study are explained.

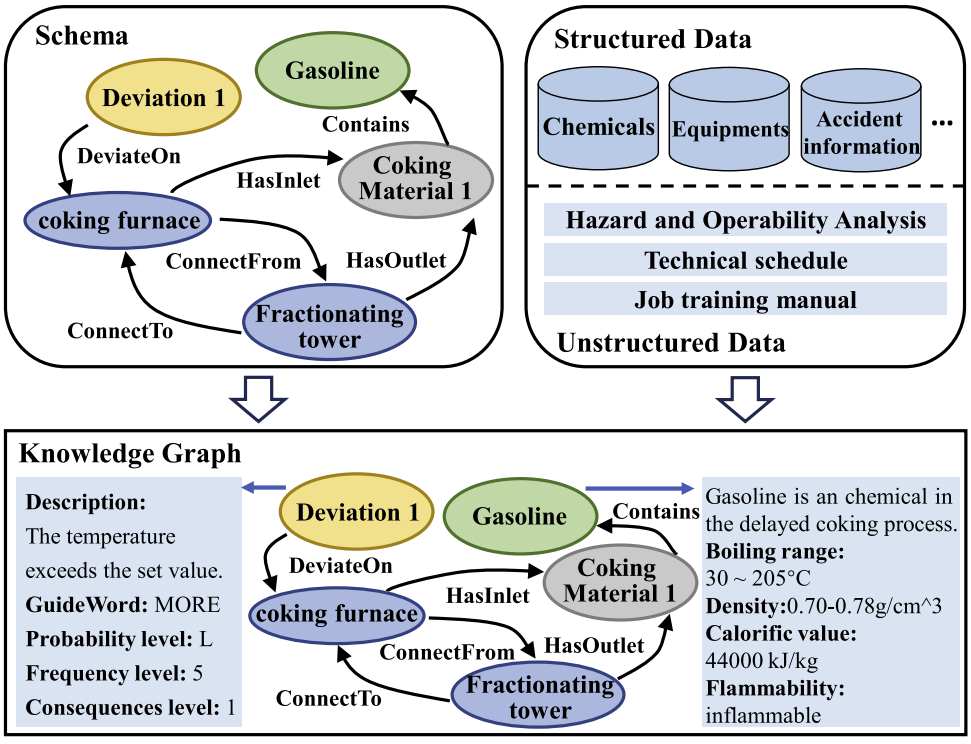


Fig. 5. Construction of process safety knowledge graph.

3.1. Delayed coking process

Delayed coking is a high-strength thermal cracking process for heavy oil processing. The process unit thermally decomposes petroleum residuum, which are typically the bottoms from the atmospheric and vacuum distillation of crude oil to produce a value-added mixture composed of olefins, naphtha, gaseous oil and petroleum coke. The flexibility of the delayed coking process can handle different raw materials, which will provide a solution for the refinery to solve the problem of reduced remaining fuel demand and use economic advantages to upgrade it to more valuable light products. Conventional methods such as coking, visbreaking, vacuum distillation, solvent deasphalting or residual cracking have affected the upgrading of residual raw materials, resulting in less residual fuel. However, in addition to coking, they still produce liquid bottom products that require canning and cutting materials. Therefore, the delayed coking process will no longer produce residual oil, which illustrates the importance of delayed coking in the coking unit and also reflects one of the main advantages of the delayed coking process.

However, the entire reaction in delayed coking process is endothermic and requires a lot of high-grade energy. The demand for high temperature and high pressure makes the delayed coking process a relatively dangerous production process, which makes it very important to monitor and ensure the operating status of the entire delayed coking process. Thus, in this paper, the delayed coking process is chosen as a case to develop a knowledge graph.

3.2. Ontology definition

For process safety in delayed coking, there are 10 entity types, 16 relationships and 30 entity attributes defined in a mixed manner, which is proposed in the ONTOLOGY DESIGN section. As shown in Tables 1 and 2, a simple schema definition for the delayed coking process is given for convenience, including some entity types, relationships, and entity attributes.

Table 1
Definition of entity types and relationships.

Entity types	Relationships
ProcessEquipment	ConnectTo
ProcessFlow	ConnectFrom
Chemicals	DeviateOn

Table 2
Definition of entity attributes.

	Entity attributes
Attributes of ProcessEquipment	Attr-TagID-ProcessEquipment
	Attr-SetPressure-ProcessEquipment
Attributes of Chemicals	Attr-BoilingRange-chemical
	Attr-Flammability-chemical

To explain the defined schema for the process safety in the delayed coking, an example is shown in Fig. 6. This figure contains various entity types, such as ProcessEquipment, Deviation, ProcessFlow, and Chemical. The concepts of process safety related to these entity types are provided in Fig. 6. For example, based on the pipe connections shown in CAD (Computer Aided Design) drawings, we define the relationship ConnectTo to depict the directional connection between the entity Fractionating tower and the entity Coking furnace. For the realization of reasoning, it is desirable to define an inverse of ConnectTo, i.e. ConnectFrom. The entity Coking Material 1, which belongs to the entity type ProcessFlow, is utilized to represent the flow from the fractionating tower to the coking furnace. This entity can indirectly indicate the directed connection between the fractionating tower to the coking furnace and has many parameters and attributes associated with process safety including temperature, pressure and flow rate. These attributes associated with process safety are defined as entity attributes in the schema for process safety knowledge graph.

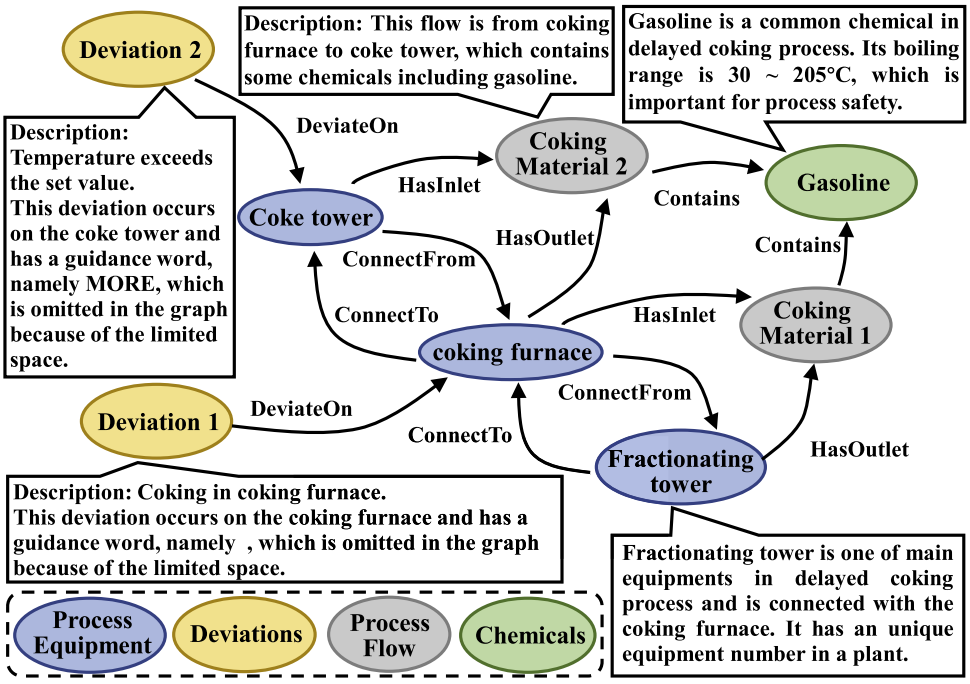


Fig. 6. An illustrative example of schema in the process safety knowledge graph.

Moreover, the entity Coking Material 1 is ensured to contain Gasoline, which is an entity belonging to the entity type Chemical. Of course, the entity Coking Material 1 also contains many other chemicals, which are omitted in Fig. 6 due to the limited space. As the most relevant component of process security, deviations are also considered and defined in the schema, such as the entity Deviation 1. Deviation 1 belongs to the designed entity type Deviations. Because Deviation 1 describes the coking phenomenon in the coking furnace, there is a relationship between entities Deviation 1 and Coking furnace, which is defined as the relationship DeviateOn. According to the description of deviation in HAZOP, the probability level, frequency level, and consequences level are defined as the attributes for the entity type Deviations.

3.3. Implementation of knowledge graph

3.3.1. Treatment of figures and diagrams

Except for data from relational databases and text documents, data in detailed process flow diagrams are also important, where there is a lot of information about the relationship between pieces of equipment. However, there is no good technique to obtain data directly from process flow diagrams. Considering the complexity and quantity of CAD drawings, manual marking is adopted as a suitable solution, in which only the relationship data between important pieces of equipment can be extracted and labeled. After manual extraction and labeling, a total of 567 relationship data is obtained.

3.3.2. Treatment of tables in HAZOP reports

Different from other documents, there are so many tables in HAZOP reports and these tables contain a wealth of data related to process safety that has been brainstormed by domain experts. To extract data from tables in HAZOP reports, our solution is depicted in Fig. 7. First, all sentences, phrases and numbers are all extracted from tables with the same structure in tables. Then, for phrases and numbers, we directly utilized Python-based programs to convert them and import them into the graph database. Different from phrases and numbers, sentences need to be processed by

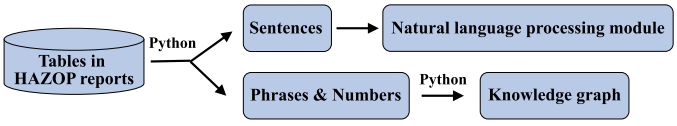


Fig. 7. A solution for tables in HAZOP reports.

natural language processing techniques and then data in sentences can be extracted. In tables of HAZOP reports, which are related to delayed coking process, we obtain 425 directly extracted entities and 522 sentences.

3.3.3. Application of natural language processing methods

For sentences in documents, natural language processing technologies are key components in the realization of data acquisition. After a comparison of training results based on the same labeled dataset, which contains more than 50,000 entities, the BERT+CRF (Bidirectional Encoder Representations from Transformers+Conditional Random Field) framework is chosen as a NER solution among technologies in natural language processing. The named entity recognition for different entities such as chemicals, organizations, deviations, and equipment has been implemented. For the obtained model, we use a manually labeled test set to evaluate its performance. The model has an accuracy rate of 0.9405, a recall rate of 0.9450, and an F1 value of 0.9427. Considering NRE, for the delayed coking process, we have constructed a dataset including 9682 relations and 3560 Chinese sentences. Similarly, after comparison, the BiLSTM+Attention (Bi-directional Long Short-Term Memory+Attention) framework is chosen as the solution of NRE (Neural Relation Extraction). The recognition and extraction of the relationship between equipment, chemicals, and logistics can be achieved with the help of manual modification. The BiLSTM+Attention network model has an accuracy rate of 0.4975, a recall rate of 0.4925, and an F1 value of 0.4950. The reason why these three indicators are not high is the complexity of the Chinese. In the case of delayed coking process, 5615 entities and 2680 relations are extracted from HAZOP reports and other documents.

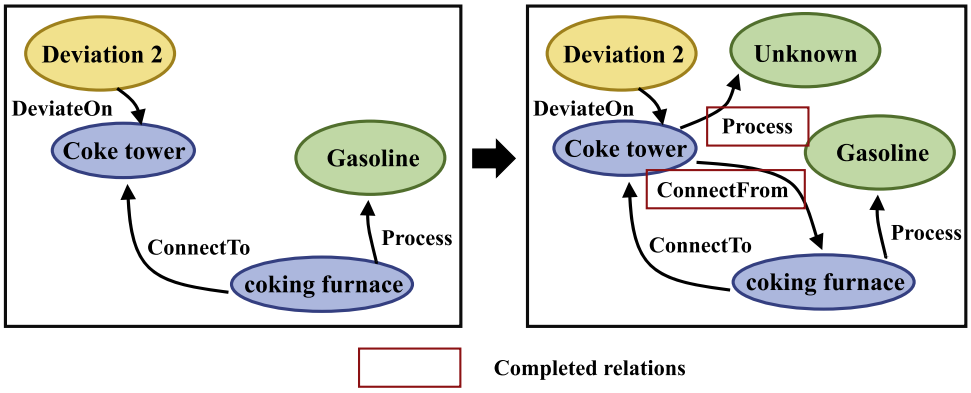


Fig. 8. An example for relation completion.

4. Discussion

4.1. Relation completion

As we all know, HAZOP reports are realized in the form of brainstorming with the rules and knowledge from experts. Thus, it is found that there are always some relations between different risk factors that are missed in the HAZOP reports. Based on the designed rules in the constructed ontology, we can easily find some missed relations in the constructed knowledge graph. An example is provided in Fig. 8. The designed rules in the constructed ontology mainly includes 'Functional', 'Inverse functional' and 'Transitive', which refer to OWL[4] rules. In the future, we want to further find missed relations between risk factors according to chemical principles and hiding rules contained in operating data.

4.2. Applications related to hazard and operability analysis

Just like the Internet industry, the construction of the process safety knowledge graph finally lays a solid foundation for the development of subsequent applications. Various applications are used directly to improve the process safety level of delayed coking manufacturing. After consideration and consultation with the domain experts, three main applications are chosen to be realized, including visualization, information retrieval, and recommendation. For convenience, these three applications are embedded into a service platform.

(1) Knowledge visualization

For operators and the general public, the graphical representation of knowledge contained in the process safety knowledge graph is friendliest because it minimizes the complexity of the entities and exhibits relationships clearly. To realize visualization, D3.js, which is a JavaScript library, is chosen to be a solution for visualization. The visualization interface for the process safety knowledge graph is shown in Fig. 9. In Fig. 9, the service platform graphically shows different brief conceptual information related to the coke tower, including deviations and flows. Bow-tie methods can also make a visualization for events related to process safety. However, only threats and consequences of events are provided in bow-tie methods. In our knowledge visualization, other information, such as designed parameters and recommendations, can also be provided based on the constructed knowledge graph. Thus, operators can select any of these entities to further explore the attribute information. This service platform can help operators to better understand the knowledge of the process safety, which may effectively reduce the accidents caused by operators and improve the process safety level.

(2) Automatic information retrieval

Based on the realized visualization, the effectiveness of information retrieval can be greatly improved. Different from the text-matching retrieval, the information retrieval based on the knowledge graph focuses on the entities. As shown in Fig. 9, when we search the fractionating tower, its attributes and all entities related to it are provided. Information retrieval is the main foundation of the question-answering function, which is our future tasks.

(3) Extraction of causal chains

When an accident occurs, operators need to obtain the most reasonable operation solution. Because the manufacturing process is too complicated, it is difficult for operators to obtain a reasonable operating solution without fully understanding the process technology. With the constructed process safety knowledge graph and information retrieval function, the automatic operation solution recommendation function can be realized. The implementation idea is shown in Fig. 10. For an accident, all potential causes can be retrieved via the information retrieval function. By comparing the probability levels of all potential causes, all the corresponding operation solutions can be sorted. Then the top three solutions are recommended to the operators.

Some existing methods, such as fault trees, event trees and Bayesian networks, are also able to realize the causal chains extraction. However, these methods can not be directly applied on the constructed graph and need extra data, such as dependent probabilities. With probabilistic data, these methods can realize inference analysis based on quantitative calculation. Different from fault trees, event trees and Bayesian networks, our extraction of causal chains fully depends on the constructed knowledge graph and does not need extra data. Besides, based on relation completion for knowledge graph, extraction of causal chains in this paper can be complemented. Of course, based on the graph construction of knowledge graph, qualitative inference analysis of our extraction of causal chains can be realized. Due to the qualitative characteristic, more hidden causal chains may be found.

4.3. Relationship with HAZOP-based expert system

Based on HAZOP reports, some expert systems have been designed and realized, including optHAZOP (Khan and Abbasi, 1997), ExpHAZOP$^+$ (Rahman et al., 2009) and LDGHAZOP (Cui et al., 2010). These HAZOP-based expert systems aim to semi-automatically construct HAZOP reports with rules from experts and manual correction of users. Different from HAZOP-based expert systems, the construction solution of process safety orientated knowledge graphs in this paper aims to utilize HAZOP reports to

[4] w3.org/2001/sw/wiki/OWL.

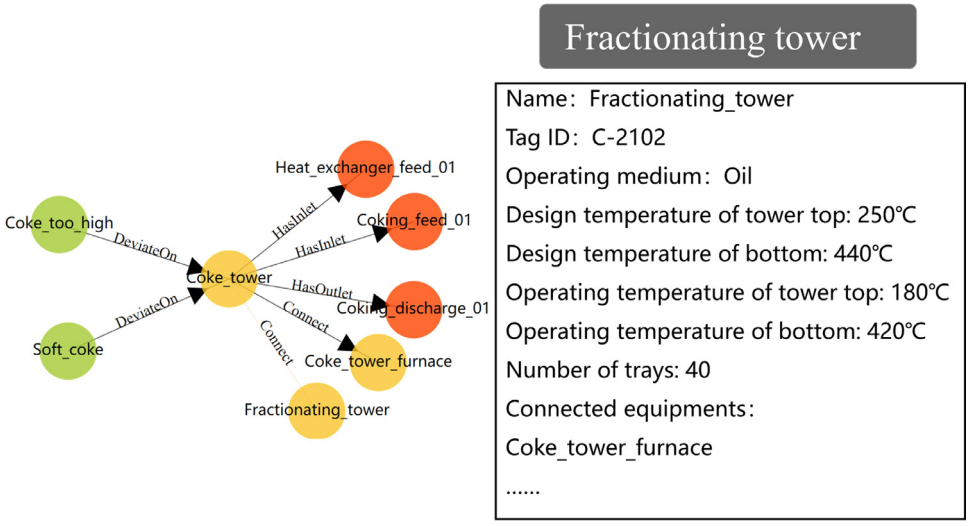


Fig. 9. Knowledge visualization for the coke tower and automatic information retrieval for the fractionating tower.

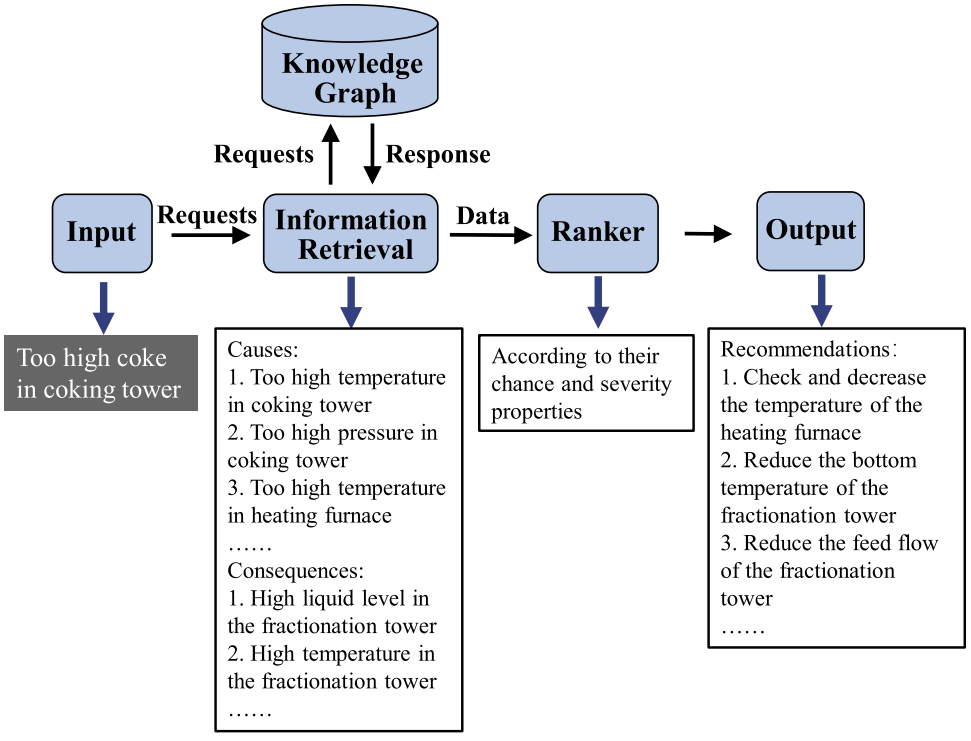


Fig. 10. Implementation idea of recommendation function.

provide a foundation for improving process safety. The construction solution presents the causal relationship analysis contained in the HAZOP report in the form of a semantic network, which can help assist emergency response, build a recommendation system and assist operator training.

4.4. Implementation in the practical industry

In a factory, except for static knowledge data, dynamic operational status and monitoring data is also needed to be integrated into the process safety knowledge graph, which is the problem we're going to solve. When the above problem has been solved, those applications proposed in the subsection entitled Applications related to Hazard and Operability Analysis can be directly applied in a factory. The realization of the knowledge graph construction and related applications in an industrial setting only needs some hardwares including servers and High-performance GPU (Graphics Processing Units) when the ontology has been designed. The ontology design step requires the participation of some experts and the support of factory data.

5. Conclusions

As an emerging technology in knowledge representation and management, the knowledge graph is applied for the process safety in the chemical industry in this paper. We propose a comprehensive semi-automatic construction framework including ontology definition, data acquisition, import and storage. In this pa-

per, delayed coking, which is an important process in the oil refining industry, is chosen as a case to develop a knowledge graph to improve process safety level. The construction framework makes full use of the knowledge of delayed coking process and provides the foundation for improving process safety level. Then, based on the constructed knowledge graph, some applications closely related to process safety are also developed. These applications that rely on the constructed knowledge graph can provide a novel solution to realize process safety monitoring, consequence analysis, accident tracing, and other safety functions, which are important to improve the level of the process safety in the chemical industry.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Shuai Mao: Methodology, Software, Writing - original draft, Visualization. **Yunmeng Zhao:** Writing - original draft, Methodology. **Jinhe Chen:** Resources. **Bing Wang:** Conceptualization, Writing - review & editing, Supervision. **Yang Tang:** Conceptualization, Supervision, Project administration.

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